User Documentation

Overview

Guide for using RTDIP Forecasting Components to analyze and forecast Shell sensor data using machine learning models.

Prerequisites

Dev container built and running (VS Code Dev Containers)
Conda environment activated: conda activate rtdip-sdk
Azure credentials configured for data ingestion from Azure Blob Storage

Quick Start

```
cd amos_team_resources/shell
python pipeline_shell_data.py
```

This invokes the AMOS-added workflow:

- Data Ingestion - Download data from Azure Blob Storage
- Preprocessing - Outlier removal, data cleaning, feature extraction
- Flat Sensor Detection - Identify sensors with minimal variation
- Flat Sensor Filtering - Remove uninformative sensors
- Anomaly detection – Identify anomalies in the data
- Decomposition – Apply decomposition to the data
- Model Training - Train AutoGluon ensemble
- Model Optimization - Tag best model for deployment
- Visualization - Generate forecast plots

Pipeline Options:

- Full pipeline with data ingestion

```
export AZURE_ACCOUNT_URL="https://<placeholder>.blob.core.windows.net"
export AZURE_CONTAINER_NAME="<placeholder>"
export AZURE_SAS_TOKEN="?sv=..."
python pipeline_shell_data.py --ingest-format parquet
```

- Skip specific steps (if data already exists)

```
python pipeline_shell_data.py --skip-ingestion
python pipeline_shell_data.py --skip-preprocessing
python pipeline_shell_data.py --skip-flat-detection
python pipeline_shell_data.py --skip-flat-filtering
python pipeline_shell_data.py --skip-training
python pipeline_shell_data.py --skip-visualization
```

- Custom data paths

```
python pipeline_shell_data.py --raw-data data/ShellData.parquet
--preprocessed-data output.parquet
```

- Use sample of data (for testing/memory-constrained systems)

```
python pipeline_shell_data.py --sample 0.1
```

- Use Plotly for interactive HTML visualizations

```
python pipeline_shell_data.py --use-plotly
```

- Customize flat sensor detection thresholds

```
python pipeline_shell_data.py --std-threshold 0.05 --unique-ratio-
threshold 0.02
```

- Customize outlier detection

```
python pipeline_shell_data.py --n-sigma 5.0
```

Basic Workflow

RTDIP x AMOS

Training Scripts